ModelIt!

Teacher's Guide

# Can Nanotechnology Deliver Medicine to Individual Cells?

Modeling Targeted Drug Delivery, Nanoparticle Design, and Therapeutic Precision

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 11L3 |
| **NGSS Standard** | HS-LS1-3, HS-PS2-4 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: HS-LS1-3, HS-PS2-4**

Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis. Use mathematical representations of Newton's Law of Gravitation and Coulomb's Law to describe and predict the gravitational and electrostatic forces between objects.

## Three-Dimensional Learning

| | |
|---|---|
| Science Practice | **Planning and Carrying Out Investigations** — Students plan investigations to test how nanoparticle design parameters affect biodistribution and therapeutic efficacy, controlling variables like particle size, surface chemistry, and targeting ligand density. |
| Disciplinary Core Idea | **LS1.A Structure and Function / PS2.B Types of Interactions** — Students model how nanoparticle surface chemistry determines interactions with biological systems — electrostatic forces, receptor-ligand binding, and protein adsorption — and how these nanoscale interactions determine macroscale therapeutic outcomes. |
| Crosscutting Concept | **Scale, Proportion, and Quantity** — Students analyze how the nanoscale properties of drug delivery particles — size, surface area, charge — create emergent behaviors at the biological scale that differ fundamentally from bulk material properties, and use quantitative reasoning to evaluate the 0.7% delivery efficiency in context. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A patient receives an injection of drug-loaded nanoparticles with a circulation half-life of 4 hours. After 12 hours (three half-lives), what percentage of the original nanoparticle dose remains in circulation, and if the tumor accumulation rate is 0.1% of circulating dose per hour, what is the approximate total tumor accumulation over this period?

**CONSTRUCTED RESPONSE:**

Using evidence from your model, explain the PEGylation dilemma in nanoparticle design: why dense PEG coating is necessary for survival in the bloodstream but counterproductive for drug delivery at the target site. Propose and evaluate one design strategy to resolve this trade-off, referencing at least two model variables in your response.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Conventional chemotherapy is a blunt instrument: toxic drugs are injected into the bloodstream and distribute throughout the entire body, killing rapidly dividing cells wherever they are found — cancer cells in the tumor, but also hair follicle cells, gut lining cells, and bone marrow cells, causing the devastating side effects that make chemotherapy so brutal. The fundamental problem is targeting: how do you get the drug to the cancer and nowhere else? Nanotechnology offers a potential solution by packaging drugs inside nanoscale carriers designed to navigate biological barriers, evade immune destruction, and concentrate their payload at the tumor site. The vision is compelling. The reality is more complex — and more instructive — than the headlines suggest.

## Nanoparticles as Drug Carriers

Drug-loaded nanoparticles range from 10 to 200 nanometers in diameter — roughly 100 to 1,000 times smaller than a human cell. They can be made from lipids (liposomes, the most clinically successful platform), polymers (PLGA, PLA, chitosan), proteins (albumin), or inorganic materials (gold, silica, iron oxide). The drug is either encapsulated inside the particle, embedded in the particle matrix, or attached to the particle surface. Nanoparticle encapsulation solves several pharmaceutical problems simultaneously: it protects fragile drugs from degradation in the bloodstream, makes insoluble drugs injectable by packaging them in water-compatible carriers, and prevents the drug from interacting with the body until it is released at the intended site. The first FDA-approved nanomedicine — Doxil, a liposomal formulation of doxorubicin — was approved in 1995 and demonstrated that nanoparticle encapsulation could reduce the severe cardiac toxicity of free doxorubicin by altering the drug's biodistribution.

## Biological Barriers to Nanoparticle Delivery

The journey from injection site to target cell is an obstacle course. First, blood proteins immediately adsorb onto the nanoparticle surface (the protein corona), changing its biological identity and often marking it for immune destruction. Macrophages in the liver (Kupffer cells) and spleen recognize opsonized particles and engulf them within minutes — this is why 30-99% of most nanoparticle doses end up in the liver. Particles that survive immune clearance must find and enter the target tissue. For tumors, the enhanced permeability and retention (EPR) effect provides a passive mechanism: tumor blood vessels have gaps of 200-800 nanometers (normal vessels have gaps of less than 2 nanometers), allowing appropriately sized nanoparticles to leak into tumor tissue. However, the EPR effect varies enormously between tumor types, between patients, and even within a single tumor. After entering the tumor, particles must penetrate through the dense extracellular matrix to reach cancer cells, bind to or be internalized by the cells, and release their drug cargo at a therapeutic concentration.

## PEGylation and the Stealth Challenge

Polyethylene glycol (PEG) coating — PEGylation — revolutionized nanomedicine by creating a hydrophilic brush layer that reduces protein adsorption and immune recognition. PEGylated nanoparticles circulate for hours instead of minutes, dramatically increasing the opportunity for tumor accumulation. However, PEGylation creates a dilemma: the same polymer brush that shields the particle from macrophages also physically blocks targeting ligands from reaching cell surface receptors and can inhibit the cellular uptake needed for drug delivery. Additionally, repeated PEGylated nanoparticle injections can trigger anti-PEG antibodies — the accelerated blood clearance (ABC) phenomenon — where subsequent doses are cleared faster than the first. Researchers are exploring alternatives to PEG including zwitterionic polymers, cell membrane coatings, and CD47 'do not eat me' peptides that mimic the signals human cells use to avoid immune attack.

## The Targeting Reality Check

A landmark 2016 meta-analysis by Wilhelm et al. in Nature Reviews Materials analyzed data from over 200 published studies and found that the median percentage of injected nanoparticle dose reaching tumors was only 0.7% — meaning 99.3% of injected nanoparticles went elsewhere. Active targeting (adding ligands) improved this to approximately 0.9% — a statistically significant but practically modest improvement. These numbers shocked the nanomedicine community and sparked intense debate about whether the field's fundamental approach is viable. Defenders note that even 0.7% delivery can be clinically

meaningful if it concentrates the drug at the tumor while reducing exposure elsewhere, and that the comparison should be against conventional drugs (which also have low tumor delivery) rather than against the theoretical ideal of 100% targeting. Critics argue that after 30 years and billions of dollars of research, the field needs to fundamentally rethink its approach. The debate continues, and the honest answer is that both sides have valid points.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**

Students identify seven key components of a nanoparticle drug delivery system: nanoparticle size, surface PEGylation, targeting ligand density, immune clearance rate, tumor accumulation, drug release rate, and off-target toxicity.

**E** **Establish Relationships**

Students map relationships showing that nanoparticle size and PEGylation are the primary controllable variables determining immune clearance rate, which combined with the EPR effect determines tumor accumulation, while targeting ligands improve cellular uptake specificity and drug release rate determines therapeutic concentration.

**V** **Visualize & Model**

Students build a computational model showing the seven-component nanomedicine system with drug biodistribution as the key dependent outcome.

**E** **Evaluate Outcomes**

Students run passive targeting, active targeting, and stimulus-responsive release scenarios to discover the realistic performance envelope of nanoparticle drug delivery and the biological barriers that limit it.

**R** **Revise & Extend**

Students add blood-brain barrier penetration, combination payload, or theranostic capability to model more advanced nanomedicine applications and their additional design constraints.

### Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes. |
| **As a "what-if" tool** | When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Chemotherapy works by poisoning every fast-growing cell in your body and hoping the cancer dies before you do. What if we could design a medicine so precise that it finds individual cancer cells, delivers a lethal dose of drug directly to them, and leaves every healthy cell untouched? That is the promise of nanomedicine. The biology makes it much harder than it sounds."*

**Watch For**

Open with the contrast between chemotherapy's blunt approach and nanomedicine's precision vision. Ask: if you could design the perfect drug delivery system, what properties would it have? List them. Then reveal the biological barriers.

## Slide 2: Learning Objectives

3 min

**Say:** *"Today you are modeling the journey of a drug-carrying nanoparticle from injection to tumor. You will discover why the body treats these particles as invaders, why most never reach their target, and why even imperfect delivery is still a major improvement over flooding the body with toxic drugs."*

**Watch For**

Have students read objectives. Pre-teach 'opsonization' as the body's foreign particle detection system. Pre-teach the 0.7% median delivery rate and let students react to this number before explaining its context.

## Slide 3: Big Question

3 min

**Say:** *"We CAN. We already DO — there are over a dozen FDA-approved nanomedicines. But the precision is not what the headlines suggest. Your model will show you exactly how much drug reaches the tumor, where the rest goes, and why even small improvements in targeting save lives."*

**Watch For**

Quick-write: Students estimate what percentage of an injected drug dose reaches a tumor. Most will guess 50-90%. Reveal the 0.7% reality and discuss why this is both discouraging and meaningful.

## Slide 4: LEVER Framework

2 min

**Say:** *"We are modeling a system where nanoscale engineering meets the full complexity of human biology — where the immune system is both the obstacle and the inspiration for design."*

**Watch For**

Preview LEVER steps. Emphasize that this model connects materials science (nanoparticle design) to immunology (immune clearance) to oncology (tumor biology) to pharmacology (drug delivery).

## Slide 5: Activity 1: Components

8 min

**Say:** *"Seven components. Two you design: nanoparticle size and PEG surface coating. Five that biology determines in response to your design choices. The most important insight: the body does not want foreign particles in the bloodstream, no matter how well you design them."*

**Watch For**

Guide through components. Show electron microscope images of actual nanoparticles at scale. Discuss why size and PEGylation are the primary design levers. Calculate: at 0.7% delivery, how much drug must be injected for 1 milligram to reach the tumor?

## Slide 6: Activity 2: Connections

8 min

**Say:** *"The immune system is the gatekeeper. It decides how long your nanoparticles circulate, which determines how many can reach the tumor. PEG buys time, but nothing buys permanent immunity from immune clearance. Design is about optimizing survival time, not achieving invincibility."*

**Watch For**

Help students map the immune clearance pathway as the central bottleneck. Calculate circulation half-life effects: if half-life is 4 hours, what fraction remains after 12 hours? How does this limit the delivery window?

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Three designs, each building on the last. Watch the numbers: how much drug reaches the tumor, how much reaches the liver, and how much stays in circulation."*

**Watch For**

Students predict tumor accumulation before each scenario. The passive targeting scenario establishes the baseline. The active targeting scenario shows modest improvement. The stimulus-responsive scenario demonstrates how controlled release amplifies the therapeutic advantage of even modest targeting.

## Slide 8: Discoveries

5 min

**Say:** *"Nanomedicine's real advantage is not targeting precision — it is therapeutic index improvement. Even at 0.7% tumor delivery, the nanoparticle formulation concentrates drug at the tumor relative to the heart, kidneys, and gut,*

*reducing the side effects that limit how much conventional chemotherapy patients can tolerate."*

**Watch For**

Lead discussion reframing nanomedicine from the targeting precision narrative to the therapeutic index narrative. Compare side effect profiles: conventional doxorubicin causes heart damage, Doxil liposomal formulation does not. Same drug, different distribution, better outcomes.

## Slide 9: STEM Challenge

12 min

**Say:** *"A real oncology lab needs your nanoparticle design for a specific cancer. Choose your cancer type, design the particle, and show the numbers."*

**Watch For**

Groups choose between brain, pancreatic, or melanoma cancer targets and design complete nanoparticle delivery systems. Must specify particle architecture, targeting strategy, release mechanism, predicted biodistribution, and honest comparison to conventional therapy. Present with model evidence.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Nanoparticle Size, Surface PEGylation
- **INTERNAL:** Targeting Ligand Density, Immune Clearance Rate, Tumor Accumulation, Drug Release Rate, Off-Target Toxicity

## Relationships

- **Surface PEGylation to Immune Clearance Rate:** NEGATIVE (-) — Denser PEG coating creates a more effective stealth layer that reduces protein adsorption and macrophage recognition, directly decreasing the rate at which the immune system removes nanoparticles from circulation. This extended circulation time is the most important effect of PEGylation.
- **Immune Clearance Rate to Tumor Accumulation:** NEGATIVE (-) — Faster immune clearance removes nanoparticles from the bloodstream before they can accumulate at the tumor site through the EPR effect. Each pass through the liver removes a fraction of circulating particles, so faster clearance means fewer particles remain available for tumor delivery.
- **Nanoparticle Size to Tumor Accumulation:** NONLINEAR (optimal range) — Particles must be large enough (greater than 10 nanometers) to avoid rapid kidney filtration and small enough (less than 200 nanometers) to exploit the enhanced permeability and retention effect through tumor vessel gaps. The optimal size range for tumor accumulation is typically 50-150 nanometers, with accumulation dropping off sharply outside this window.
- **Drug Release Rate to Off-Target Toxicity:** POSITIVE (+) for premature release — If drug is released too quickly — before the nanoparticle reaches the tumor — the freed drug distributes throughout the body causing off-target toxicity similar to conventional chemotherapy. Slower, controlled release that activates preferentially at the tumor site reduces off-target exposure and toxicity.

## Key Discoveries

- The enhanced permeability and retention effect is real but modest — tumor blood vessel leakiness allows some nanoparticle accumulation, but the typical 0.7% tumor delivery rate means that 99.3% of the injected dose goes elsewhere, primarily to the liver and spleen where macrophages aggressively clear foreign particles
- PEGylation creates a fundamental trade-off: the stealth coating that protects nanoparticles from immune clearance also reduces their ability to be taken up by target cells, because the same polymer chains that hide the particle from macrophages also physically block targeting ligands from reaching cell surface receptors
- Active targeting with ligands improves the specificity of cellular uptake but does not dramatically increase the total amount of nanoparticles reaching the tumor — the biological barriers of immune clearance and liver sequestration remove most particles before they encounter any target cells
- The therapeutic advantage of nanomedicine over conventional drugs comes less from dramatically improved targeting and more from altered biodistribution, controlled release kinetics, and the ability to deliver drugs that would otherwise be insoluble, unstable, or too toxic in free form

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Nanoparticles deliver drugs directly to cancer cells like guided missiles" | The guided missile metaphor is the most common and most misleading description of nanomedicine. In reality, nanoparticles are more like messages in bottles thrown into a river — most wash up on the wrong shore (the liver), some are destroyed by the current (immune clearance), and a tiny fraction (0.7%) drift to the intended beach (the tumor). Active targeting with ligands improves the chances that particles reaching the tumor enter cancer cells rather than just sitting in the tumor tissue, but it does not dramatically improve the fraction that reaches the tumor in the first place. The model quantitatively demonstrates this: the major loss pathway is immune clearance before tumor arrival, not failure to enter cancer cells after tumor arrival. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Nanotechnology in medicine is futuristic and decades away from clinical use" | Nanomedicine is already here: over a dozen FDA-approved nanoparticle drug formulations are in routine clinical use. Doxil (liposomal doxorubicin) was approved in 1995 — thirty years ago. Abraxane (albumin-bound paclitaxel) is a standard treatment for breast and pancreatic cancer. The mRNA vaccines from Pfizer-BioNTech and Moderna that were administered to billions of people during the COVID-19 pandemic use lipid nanoparticle delivery — making LNPs the most widely administered nanomedicine in history. The model helps students distinguish between current clinical reality (imperfect but useful nanoparticle formulations) and future aspirations (precise single-cell targeting). |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Bigger nanoparticles can carry more drug so they must be better" | Larger nanoparticles do have higher drug loading capacity, but size affects every biological interaction the particle encounters. Particles larger than 200 nanometers cannot exploit the EPR effect because they do not fit through tumor blood vessel gaps. Particles larger than 150 nanometers are cleared more rapidly by the liver and spleen. Particles larger than 5,000 nanometers can block capillaries, causing potentially fatal embolism. The optimal size for cancer nanomedicine is typically 50-150 nanometers — a compromise between drug loading, tumor penetration, immune evasion, and tissue distribution. The model demonstrates this nonlinear size-performance relationship and shows why the optimal size is a narrow window rather than simply bigger is better. |